

Shelden Rattray

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EDUCATION

Georgia State University (Honors College) *Computer Science, B.S.*

Expected May 2027

Awards \$2500 Delta Analytics Scholarship Recipient, 5x President's List Recipient

GPA: 4.00/4.00

Activities: STEM Tutor for undergraduate CS course curriculum.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C#, Dart, SQL, Bash, HTML/CSS

Cloud & DevOps: AWS, Azure (AZ-900), GCP, Kubernetes, Terraform, Docker, CI/CD, Git, GitHub

Frameworks & Tools: React, Spring Boot, Redis, Firebase, PostgreSQL, TensorFlow, PyTorch, NumPy, Pandas

EXPERIENCE

Cloud Platform Engineering Intern | Fiserv (Fortune 500)

May 2026 – Present

- Architected and modularized infrastructure-as-code (IaC) using **Terraform** to automate the provisioning of highly available multi-region architectures across **AWS** and **Azure**.
- Orchestrated containerized workloads using **Kubernetes (K8s)** and Docker, configuring auto-scaling policies and resource limits to optimize cluster efficiency and ensure zero-downtime deployments.
- Designed and implemented high-performance caching strategies using **Redis Cache**, drastically reducing database read latencies and enhancing application throughput under peak traffic loads.
- Engineered robust **CI/CD pipelines** paired with custom **Bash/Python scripts** to automate system maintenance, VM patch management, and configuration compliance across hybrid cloud environments.

Software Engineering Intern | CuesHub

Jul 2025 – Aug 2025

- Built 10+ interfaces in for CuesHub mobile application for real-time heart rate and accelerometer visualizations.
- Implemented in-app logging utilities and structured exception handling to improve issue traceability and application stability.
- Optimized state management and caching mechanisms for 30% improvement in rendering performance.

PROJECTS

Polysia | Full-Stack Web Application | React, Firebase, PostgreSQL, RAG – [Website](#) – [GitHub](#)

- Built and deployed a full-stack web application that tracks, creates and reads stories in Mandarin Chinese to assist language learners.
- Implemented a simple model that takes user character retention data via a simple SRS-based learning system and maps the data into a custom Adaptive Learning Model, accurately determining each user's learning rate.
- Integrated LLM-based conversational feedback pipelines powered by DeepSeek and Azure Neural TTS.

Timeout Optimization Model | Python, XGBoost, React | Hacklytics 2026 – [GitHub](#)

- Developed a hybrid ensemble AI platform to quantify NBA momentum shifts and predict timeout effectiveness, utilizing 6 seasons of play-by-play data via the NBA API.
- Implemented a weighted ensemble algorithm to synthesize model outputs, identifying that timeouts are statistically most effective during 7-9 point runs rather than later 12-0 runs.
- Utilized Pandas and NumPy for large-scale data manipulation and normalization, optimizing model training for use in high-stakes, fast-paced game environments.

Rhythm Game Performance Calculator | Browser Extension | JavaScript – [GitHub](#)

- Developed a browser extension with 4,000+ installs, featuring a custom API integration layer for fetching, normalizing, and processing game data at scale.
- Optimized asynchronous workflows and caching to reduce load times and ensure reliable data accuracy.

LEADERSHIP & OTHER EXPERIENCE

Hackathon | 1st Place | Truist & Deloitte CTF Hackathon

- Captured 10/10 flags across real-world cybersecurity challenges on the HackTheBox's Capture The Flag platform.

Colorstack at GSU | Technical Director

- Architected and deployed the core web applications and underlying infrastructure supporting our digital initiatives.
- Engineered the official platform and technical infrastructure for PeachHacks, ensuring high availability and seamless attendee registration.